

NAM APRAMYCIN 552 IU/MG SOLUBLE POWDER

Nam Apramycin 552 IU/mg soluble powder appears to be a light brown powder.

Composition

Each mg contains

Apramycin (as apramycin sulfate) 552IU

Pharmacodynamics

Apramycin is a bactericidal aminoglycoside antibacterial, whose action results from bonding on the 30S subunit of the ribosome, preventing protein synthesis and disturbing the membrane permeability of bacteria.

Apramycin is effective against Gram-negative bacteria (*Salmonella* and *Escherichia coli*). Resistance mechanisms: Different aminoglycoside 3-N acetyltransferase enzymes (AAC-3) have been related with resistance to apramycin. These enzymes confer different cross-resistance against other aminoglycosides. Some strains of *Salmonella* Typhimurium DT104 in addition to resistance against beta-lactams, streptomycin, tetracyclines and sulphonamides carry a conjugative resistance plasmid against apramycin. Apramycin resistance can be influenced by co-selection (resistance to apramycin has been described to be located in the same mobile genetic element that other resistant determinants in Enterobacteriaceae) and cross resistance (e. g. with gentamicin).

Resistance developed by chromosomal resistance is minimal for most of the aminoglycosides.

Pharmacokinetics

The oral administration of apramycin is intended for antimicrobial activity within the gut; apramycin is poorly absorbed, but absorption may be increased in young animals and in animals with disrupted intestinal barrier.

Absorption:

Absorption may be high in new-born animals but rapidly decreases in the first weeks of life.

Calves. Serum levels peak at approximately 6 hours with a value of 2.4 µg/ml following oral administration of 40 mg apramycin/kg of bodyweight.

Distribution, biotransformation and excretion:

Apramycin is mainly excreted through faeces, under active form, and only a small quantity is excreted in the urine.

Pigs. Very little metabolism of apramycin takes place in the animal.

Dosing 10 kg pigs with ¹⁴C apramycin resulted in approximately 83% being recovered from the faeces, and 4% from the urine, as ¹⁴C apramycin.

Indications

Pigs (weaned piglets): Treatment of bacterial enteritis caused by *Escherichia coli* susceptible to apramycin.

Pre-ruminant calves: Treatment of bacterial enteritis caused by *Escherichia coli* and clinical outbreaks due to *Salmonella enterica* subsp. *enterica* serovar Dublin (*Salmonella Dublin*) susceptible to apramycin. Treatment should be based on prior confirmation of the *Salmonella* serovars involved or at least the availability of epidemiological data confirming the presence of this serovar.

Chickens: Treatment of colibacillosis caused by *Escherichia coli* susceptible to apramycin.

Rabbits: Treatment and metaphylaxis of bacterial enteritis caused by *Escherichia coli* susceptible to apramycin. The presence of the disease in the herd must be established before the product is used.

Dosage and Administration

To be administered via drinking water. Drinking systems should be clean and free of rust to avoid reduction of activity.

In the case of calves it can be administered in milk or milk replacer.

Pigs:

Administer 12,500 IU apramycin sulfate per kilogram of bodyweight (corresponding to 22.5 mg of product/kg bw), daily for 7 consecutive days.

Calves:

Administer 40,000 IU apramycin sulfate per kilogram of bodyweight (corresponding to 72 mg of product/kg bw), daily for 5 consecutive days.

Chickens:

Administer 80,000 IU apramycin sulfate per kilogram of bodyweight (corresponding to 144 mg of product/kg bw), daily for 5 consecutive days.

Rabbits:

Administer 20,000 IU apramycin sulfate per kilogram of bodyweight (corresponding to 36 mg of product/kg bw), daily for 5 consecutive days.

The intake of medicated water depends on the clinical condition of the animals. In order to obtain the correct dose, the concentration of the veterinary medicinal product has to be adjusted accordingly.

The amount of product (mg) to be incorporated per 1 L of water or milk should be established according to the following formula:

$$\frac{\text{Dose (mg product per kg body weight per day)} \times \text{mean body weight (kg) of animals to be treated}}{\text{Average daily water intake (L/animal)}} = \text{mg product per litre of drinking water/milk}$$

To ensure a correct dosage, bodyweight should be determined as accurately as possible to avoid under-dosing. Prepare the solution with fresh tap water (or milk/milk replacer for calves) immediately before use. Medicated drinking water should be refreshed or replaced every 24 hours. Milk replacer should be prepared prior to the addition of the powder. The solution should be vigorously stirred for 5 minutes.

Medicated milk/milk replacer should be consumed immediately after preparation. Water uptake should be monitored at frequent intervals during medication. In order to ensure consumption of the medicated water, animals should not have access to other water supplies whilst being treated. After the end of the medication period, the water supply system should be cleaned appropriately to avoid intake of sub-therapeutic amounts of the active substance. If it is not possible to obtain sufficient uptake of medicated water, animals should be treated parenterally (where appropriate). The maximum solubility of the product in water and milk replacer is approximately 1000 g/L. The use of suitably calibrated weighing equipment is recommended to ensure accurate measurement of the amount of product to be administered.

Contraindications

Do not administer in case of hypersensitivity to apramycin.

Do not use in calves with functional rumen.

Do not use in animals suffering from kidney disorders.

Special warnings for each target species

None.

Special precautions for use

Special precautions for use in animals

Use of the veterinary medicinal product should be based on susceptibility testing of the bacteria isolated from the animal. If this is not possible, therapy should be based on local (regional, farm level) epidemiological information about susceptibility of the target bacteria.

Where a diagnosis of Salmonella Dublin is made on the farm, then control measures including on-going monitoring of disease status, vaccination, biosecurity and movement controls should be considered. National control programmes should be followed where available.

Use of the veterinary medicinal product deviating from the instructions given in the Summary of Product Characteristics may increase the prevalence of bacteria resistant to the apramycin and may decrease the effectiveness of treatment with aminoglycosides due to the potential for cross-resistance.

Official, national and regional antimicrobial policies should be taken into account when the veterinary medicinal product is used.

Special precautions to be taken by the person administering the medicinal product to animals:

People with known hypersensitivity to apramycin or any other aminoglycoside should avoid contact with the product.

This product may cause irritation or sensitisation after skin or eye contact or inhalation.

Avoid contact with the eyes, skin and mucous membranes and inhalation of dust while preparing the medicated water/milk.

Use personal protective equipment consisting of gloves, mask, goggles and protective clothing while handling the product.

Wash hands after use.

In case of eye contact, rinse the affected area with plenty of water. In case of skin contact, wash thoroughly with soap and water. If irritation persists, seek medical advice.

In the case of accidental ingestion, seek medical advice immediately and show the package leaflet or the label to the physician.

In case of onset of symptoms after exposure such as skin rash, seek medical advice immediately and show the package leaflet or the label to the physician. Swelling of the face, lips and eyes or difficult breathing are more serious symptoms and require urgent medical assistance.

Interactions with Other Medicaments

Aminoglycosides may have a negative influence on the kidney function. The administration of aminoglycosides to animals suffering from renal impairment or in combination with substances that also affect renal function may therefore present a risk of intoxication. Aminoglycosides may cause neuromuscular blockade. It is therefore recommended to take such an effect into account when anaesthetising treated animals.

Pregnancy and Lactation

Pigs:

The safety of the veterinary medicinal product has not been established during pregnancy and lactation in sows. Use only accordingly to the benefit-risk assessment by the responsible veterinarian.

Cattle:

The use is not intended during pregnancy or lactation.

Rabbits:

Oral doses of apramycin administered from 6th to the 18th day of pregnancy (including doses below the therapeutic doses), have shown evidence of foetotoxic effects. Do not use during pregnancy.

Chickens:

Do not use in laying hens and within 4 weeks before the onset of the laying period.

Side Effects

None known.

Overdose and Treatment

Pigs:

Pigs have been given up to nine times the recommended use level in their drinking water for 28 days with no untoward reaction.

Calves:

Calves were given apramycin in milk replacer daily for five days, at doses up to 120 mg/kg of bodyweight. There was no toxic effect.

Chicken:

There was no mortality when chickens were given a single oral dose of 1,000 mg/kg of bodyweight. Chickens were given up to 5 times the recommended level for 15 days with no untoward reaction.

Possible intoxications can be recognised by the following symptoms: soft faeces, diarrhoea, vomiting (weight loss, anorexia, and similar), renal impairment and effects on the central nervous system (reduced activity, loss of reflexes, convulsions, etc.).

Do not exceed the recommended dose.

Withdrawal Period(s)

Pigs:

Meat and offal: Zero days.

Calves:

Meat and offal: 28 days.

Chickens:

Meat and offal: Zero days.

Not for use in birds producing or intended to produce eggs for human consumption. Do not use within 4 weeks of the start of the laying period.

Rabbits:

Meat and offal: Zero days.

Storage Conditions

Store below 30°C. Protect from heat and sunlight exposure.

Keep out of reach of children/ Jauhkan daripada kanak-kanak.

Any unused product or packaging should be disposed of in accordance with local legislation.

Shelf life as packaged for sales: 2 years.

Shelf life after first opening: 7 days.

Shelf life after dilution in drinking water: 24 hours.

Shelf life after dilution in milk replacer: Use immediately. Do not store.

Packaging Available

100g

Product holder and Manufacturer

Nam Pharma Sdn. Bhd.

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